



DEPARTMENT OF PHYSICS

Advanced Laboratory Course: Particle physics

ANALYSIS OF SCINTILLATING FIBER TRACKING
SYSTEM FOR THE LHCb EXPERIMENT

Study of scintillating fibers as a sub-detector system

Lab report of:
Giulio Giamello
Cem Kurt

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Introduction

In the setting of a modern particle physics experiment, many cutting edge designs using various physical processes are utilized as detector systems. In the worlds largest circular particle accelerator experiment at CERN, there are 4 points of interest in which the oppositely accelerated proton beams are crossed at 13.4 TeV center of mass energy which creates a breeding ground for a large range of physics processes due to interactions from the sea of partons. At each point of interest (ALICE, ATLAS, LHCb and CMS), there is an intricate combined detector system set up to record the (aftermath) of these events.

Each one of these experiment points are designed to be sensitive to different specific particle physics processes and an overall classification of events. The detector subsystems that are first in line of the incoming particles from the interaction points are usually the tracker subsystems. Alongside systems like magnets, Cherenkov detectors, electromagnetic and hadronic calorimeters; the tracker systems are a vital part of a combined approach to analyze the plethora of particle physics processes. These tracker systems work by merely "noticing" and transferring the knowledge of the charged particles passing through in their proximity.

The LHCb detector when ordered from the closeness to the interaction point consists of several subsystems. Firstly, the VELO (Vertex Locator) system, which are silicon-strip detectors positioned close to the beam line to provide precise reconstruction of primary and secondary vertices. Then comes the Trigger Tracker station, a silicon microstrip detector that tracks particles before they enter the magnetic field from the massive dipole magnet which bends the paths of charged particles so that their momenta can be measured downstream. Following the magnet, the tracking stations which are composed of scintillating fibers, continue to track the charged particles as they exit the magnetic field, ensuring accurate momentum measurement. Further particle identification is done by the Ring-Imaging Cherenkov Detectors (RICH) which exploit the Cherenkov effect. Complementing these systems are the calorimeters including the muon calorimeters, which measure the energy of electrons, photons, hadrons and muons through the development of particle showers. Together, these subsystems

enable LHCb to perform precise measurements and probe fundamental questions in particle physics.[?]

In this laboratory course, we are going to study and try to understand the basics of the particular tracking system of scintillating fibers from the LHCb experiment upgrade in 2019/2020.

Chapter 1

Theory/Design

1.1 Scintillating Fiber Trackers

A scintillating fiber tracker system consists of many scintillating fiber wires glued to each other side by side and in back to back layers; at the end of the fibre which a silicone photomultiplier photon measurement component is present. This component registers the actual photons created when a charged particle interacts with the scintillating fiber at any point and manages to travel down the fiber material into the silicone photomultiplier. Scintillating fibers are first combined into mats in a hexagonal order, as can be seen in [Figure 1.1](#) after which these mats are combined to cover a large enough surface area to be used in the detector as can be seen in [Figure 1.2](#). In the second figure, the mirror put in between the two connected "dead" inner ends of the fibre without a measurement component can be seen. The purpose of this mirror is to reflect photons back into the fibre so the photons can reach the silicon photomultiplier end.

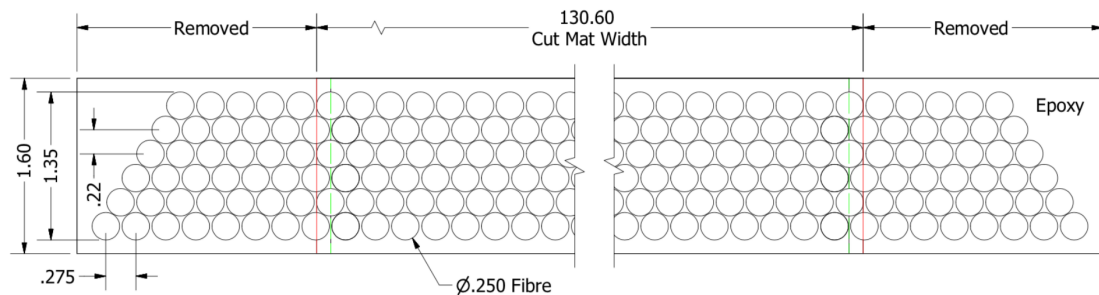


Figure 1.1: Schematic cross-section of a fibre mat with six layers of fibres arrange in a hexagonal pattern. The mat is cut along the red lines.[?]

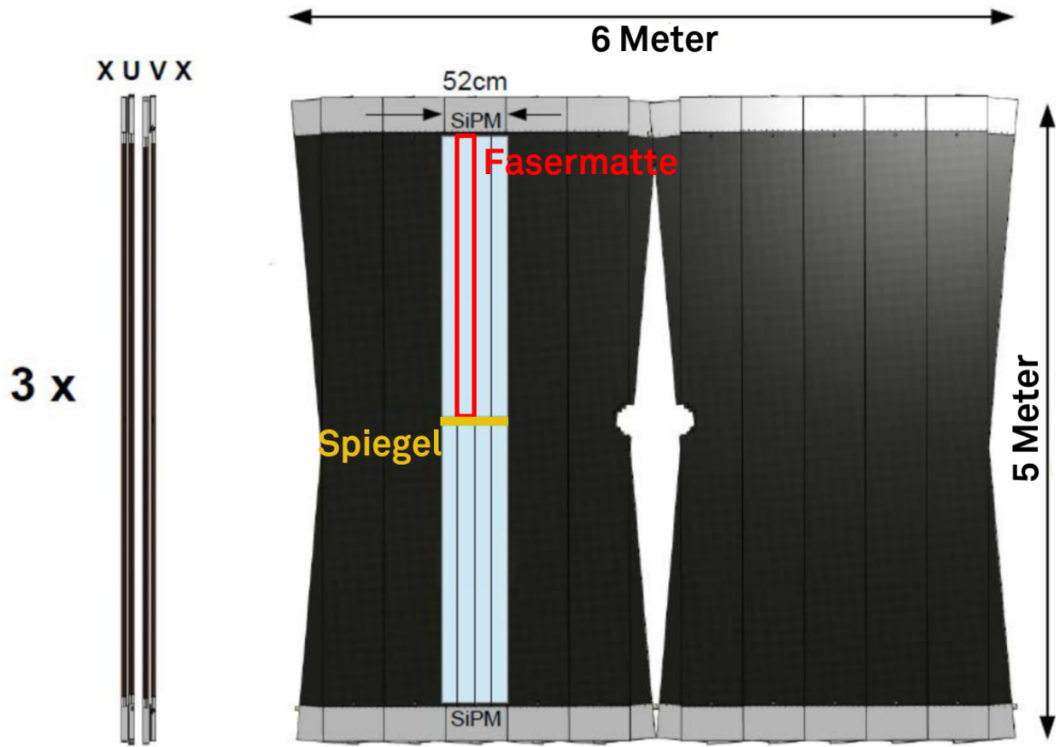


Figure 1.2: Schematic view of the SciFi Tracker. On the left side of the image is a side view of a station consisting of four layers. The right side shows a front view of the SciFi Tracker, with the two halves slightly separated. A single fibre mat is highlighted in red.[?]

The fibers in LHCb are constructed at $250 \mu\text{m}$ thickness. When a reading is measured at the photomultiplier end, the probability of the charged particle passing through a single fiber is described by a uniform distribution. The resolution of a fiber of width d can be calculated using the standard deviation of a normal distribution:

$$\sigma_{\text{fiber}} = \sqrt{\frac{d^2}{12}}.$$

Inserting the LHCb scintillating fiber width of $250 \mu\text{m}$, a resolution of $\approx 72 \mu\text{m}$ is calculated. Highly penetrating particles are scattered less in the fibers compared to other trackers, consequently the systematic error introduced by the detector is kept low.

1.2 Scintillating Fiber Structure

The fibres have an organic scintillator 220 μm thick polystyrene core, which acts as the active detector and photon propagation medium. The core is surrounded by two sheaths with outwardly decreasing refractive indices to allow an increased range of total internal reflection at the interfaces as can be seen in Figure 1.3. Each of these sheaths has a thickness of 7.5 μm , resulting in a total fiber thickness of 250 μm .

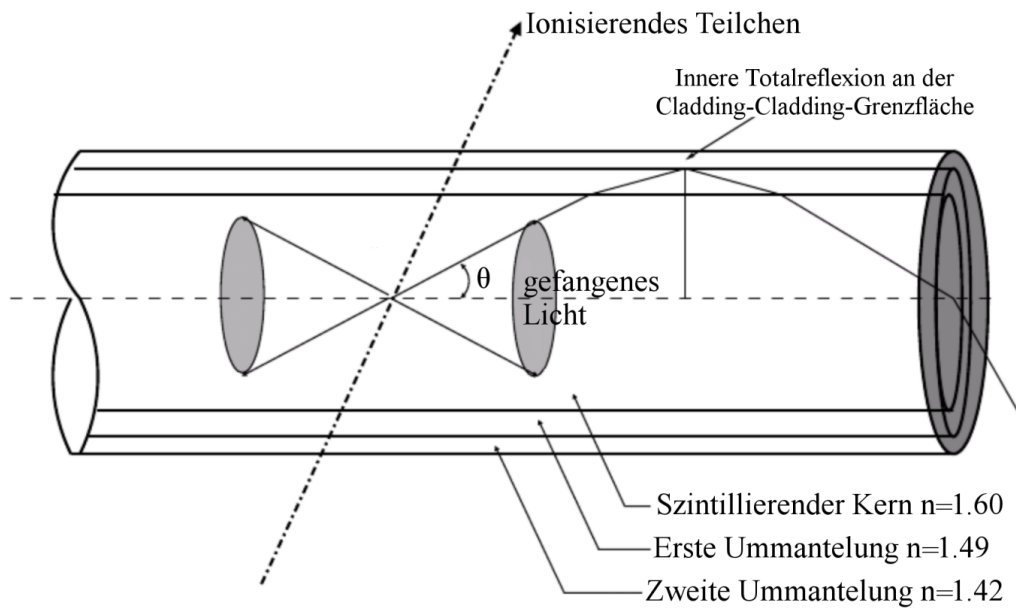


Figure 1.3: Schematic structure of a scintillating fibre of type SCSF-78MJ from the company Kuraray.[?]

When an ionizing particle passes through and interacts with the fibre, the valence electrons of the polystyrene become excited. Excitations of electrons into higher vibrational modes transition the molecule to the S_{10} energy state without radiating a photon. A photon is finally emitted when the S_{10} state transitions into the ground state S_{00} . The difference of energy in the different energy levels in the molecule is of the order of a few eV, hence the corresponding photon is emitted in the UV range.

A scintillator made up of pure polystyrene is not compatible with a particle accelerator detector however, due to its long average decay time from higher energy molecule states in the order of ≈ 300 ns. In this construction, the yield in the detector is massively decreased. To combat this, the material doped with another called "p-terphenyl" which decrease the decay time to just a few nanoseconds.

1.2.1 Propagation of Photon in the Fibers

The UV photons are created in the fiber at the excitation point, which can be anywhere within the width of the core and hence have some distance from the center of the fibre. As the photon travels down the fiber, it will reflect with a certain angle θ through one of the interfaces between the core, first sheath and the second sheath or pass through the structure and get lost. The photon that is propagating in the structure will have a minimal distance r_{min} from the center of the fiber as can be seen in Figure 1.4.

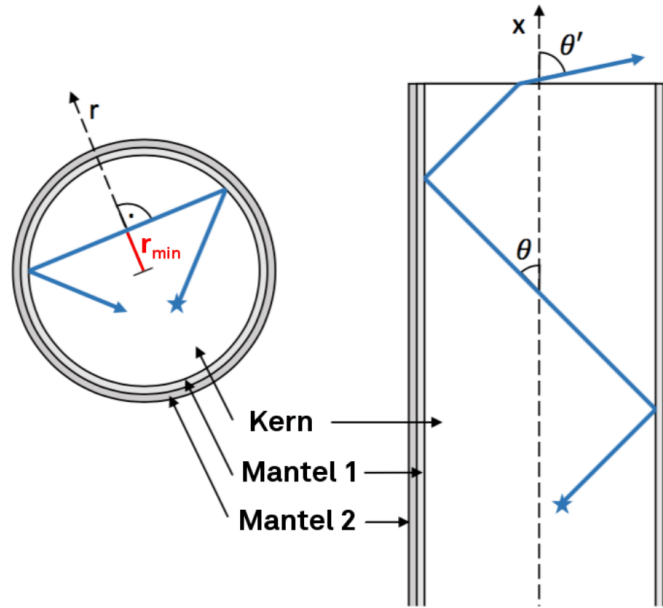


Figure 1.4: Movement of a non-meridional (non-meridional: not passing through the exact center of the core) photon through the fibre. On the left side, a cross-section through the fibre is shown. The smallest distance of the photon path to the fibre axis is denoted by r_{min} . On the right side, a side view of the same photon path is shown. The angle θ is defined as the angle to the central axis of the fibre.[?]

The length L of the path a photon takes and the number of times it reflects from the interface N are given by the equation:

$$(1.1) \quad L = \frac{x}{\cos \theta}$$

$$(1.2) \quad N = \frac{x \tan \theta}{2\sqrt{r_{core}^2 - r_{min}^2}}$$

Where, r_{core} is the radius of the fiber core and x is the distance from the end of the fiber. The reflection angle, which is the angle present at the end of the fiber θ'_{refl} can be calculated using the formula:

$$(1.3) \quad \theta'_{\text{refl}} = \arcsin \left(\sqrt{1 - \frac{r_{\text{min}}^2}{r_{\text{core}}^2}} \sin \theta \right)$$

The photons that are lost after creation obviously do not possess these specifications. There are many processes which contribute to lost photons such as Rayleigh scattering or absorption by the materials of the fiber. We find that there is a specific capture efficiency for all similar purpose materials that describes what fraction of photons will remain inside the fiber and make it to the end. In general, the attenuation I of photons after traveling a distance L in the fiber exponentially decreases according to:

$$(1.4) \quad I(x) = I_0 \exp \left(-\frac{L}{\Lambda} \right) = I_0 \exp(-aL)$$

Where Λ is the attenuation length which is calculated to be approximately 3.5 meters in the specific fibers used in this experiment. The inverse attenuation length a is defined as the attenuation coefficient. We can further distinguish the processes involved in the attenuation of the photon. We can express the losses due to the core material A_{core} and losses due to reflection A_{refl} as separate terms:

$$(1.5) \quad I(x, \theta) = I_0 A_{\text{core}}(x, \theta) A_{\text{refl}}(x, \theta)$$

Both are exponential expression and are given as:

$$(1.6) \quad A_{\text{core}}(x, \theta) = \exp \left(-\frac{L(x, \theta)}{\Lambda} \right)$$

$$(1.7) \quad A_{\text{refl}}(x, \theta) = \exp(-\varepsilon N(x, \theta))$$

Where for reflection losses, N is the number of reflections and ε is the probability of loss during reflection. The final expression for the attenuation length is then obtained as:

$$(1.8) \quad I(x, \theta) = I_0 \exp \left(-x \underbrace{\left(\frac{1}{\cos \theta} + \frac{\varepsilon \tan \theta}{2\sqrt{r_{\text{core}}^2 - r_{\text{min}}^2}} \right)}_{a_{\text{eff}}} \right)$$

Here, a_{eff} is defined as the effective attenuation coefficient and x is the distance from the fiber end where the excitation occurs.

Chapter 2

Data Analysis

2.1 Experiment Setup

The setup consists of a single fiber stretched from one end to the other and an LED mechanism that can move very close to and along the length of the fiber. At the end of the fiber is a rotating arm spectrometer that can register the light coming out of the fiber at various angles both vertical and horizontal, it can be tilted in the horizontal and vertical plane within the angle intervals $[-20^\circ, 90^\circ]$ and $[-6^\circ, 90^\circ]$, respectively. The simulation data is produced using Geant4 for a 50 fiber structure with excitations on 24 points across each fiber. No other details about Geant4 or production process of the simulation data is discussed.

2.2 Measurements

For every measurement, a dark count data where the measurements are taken while the LED is turned off is also taken. This will be integral in the analysis, where the dark count data will be subtracted from the actual counts to remove the background. The current of the LED is set to 20 mA for all measurements as well as the integration time which is set to $10000\mu\text{s}$ and number of averages which is set to 5 in all measurements.

2.2.1 Spectrometer Measurement

The first measurement is taken without giving any specific input into the machines to test the dependence of the system to the ambient light in the room. Two measurements are taken with the roomlight on and off.

2.2.2 Radial symmetry Measurement

This measurement is taken to test the radial symmetry of the spectrometer setup, the horizontal and vertical angle of the spectrometer are varied while the excitation position on the fiber from the LED is kept constant. The interval of angles used in this measurement is $[-18^\circ, 30^\circ]$ in the horizontal and $[-6^\circ, 35^\circ]$ in the vertical. 13 measurements are taken in both cases.

2.2.3 Intensity Measurement

For the last measurement, the intensity of light received at the spectrometer end depending on the distance is measured moving the LED structure on the axis of the fiber, hence moving the excitation point. The fiber is excited in a total range of 150 cm in 7.5 cm increments. For each excitation point, 10 different horizontal angles in the interval $[0^\circ, 40^\circ]$ are measured.

2.3 Measurement Analysis

2.3.1 Spectrometer Measurement

This section only relates to the emission spectrum of the LED we are using and the decision to turn off the roomlight before taking the consequent measurements. In [Figure 2.1](#), the LED signal can be picked up more clearly for the measurements where we have the roomlight turned off.

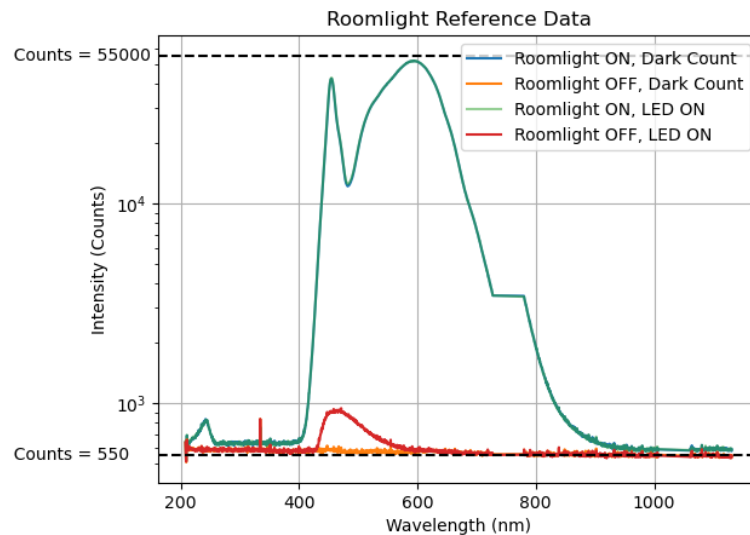


Figure 2.1: Spectrometer Measurement with Roomlight and LED toggled.

If we proceed to then subtract the dark count values where the LED is turned off from the counts where we had the LED turned on, we get the following [Figure 2.2.](#)

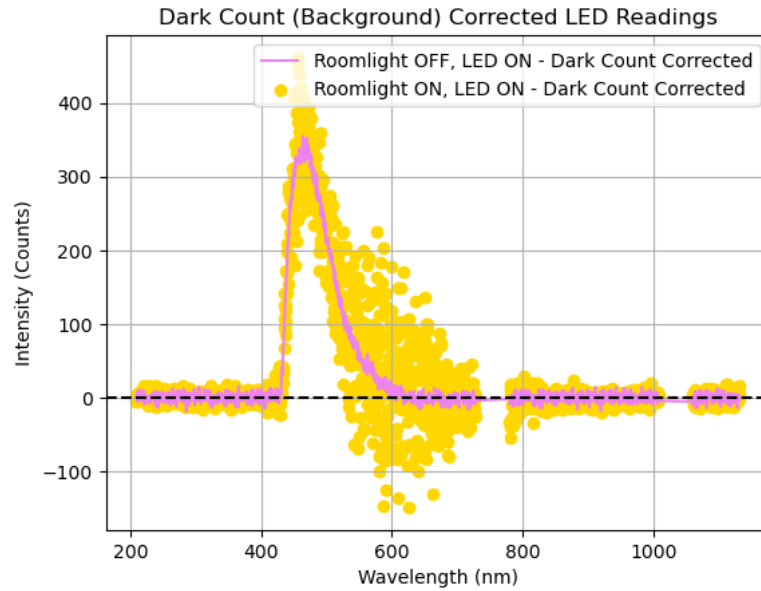


Figure 2.2: LED signal isolated graph by subtracting dark counts.

This makes the difference in the two datasets much more visible as the LED signal is smeared and has a lot of uncertainty for the dataset with the roomlight on. So the decision is made to turn off the roomlight while the rest of the measurements are taken.

2.3.2 Radial symmetry Measurement

Our 2-D histogram for total photon count plotted against varied angles of horizontal and vertical position of the spectrometer can be seen in the [Figure 2.3](#). This result uses the dark counts subtracted version for dark counts taken at each position of the spectrometer. It shows that radially, there is a symmetry in our reading and for the rest of the measurements we can make the decision to utilize only the horizontal axis to take angular dependent measurements. This is because same amount of displacement in the vertical axis would give the same data as the horizontal axis.

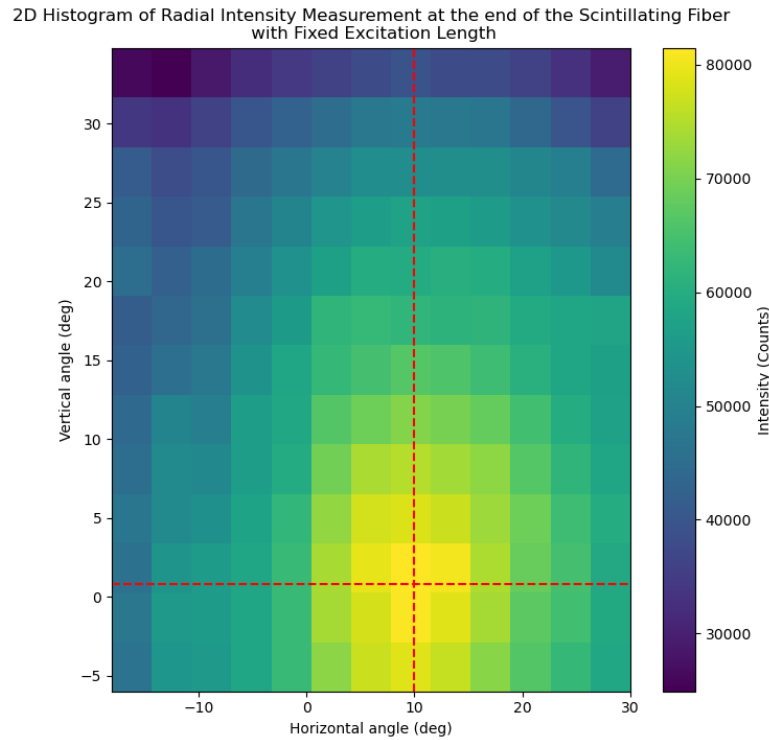


Figure 2.3: Measurement from the spectrometer angled at different points at the fiber end while the excitation point on the fiber is kept constant

2.3.3 Intensity Measurement

The final measurement gives us the data for intensity at various angles at increasingly distant excitation points. The complete data is plotted in [Figure 2.4](#). There are some obvious observations to be made such as the number of photons reaching the end of the fiber dropping as the excitation distance increases and the larger spectrometer angle measurements yielding less counts than those more inline with the fiber axis. The surprising observation here is the sudden drops of photon counts right after moving away from 100mm and 600mm excitation lengths. Our explanation for this is either a significant kink or imperfection in the fibre at those points or a disturbance of the experiment setup. We reason that these kinky regions are unrelated to our theory and to avoid them we produce decreasing power law fits for 3 intervals of excitation length without kinks: 0-75mm, 150-650mm and 725-1500mm.

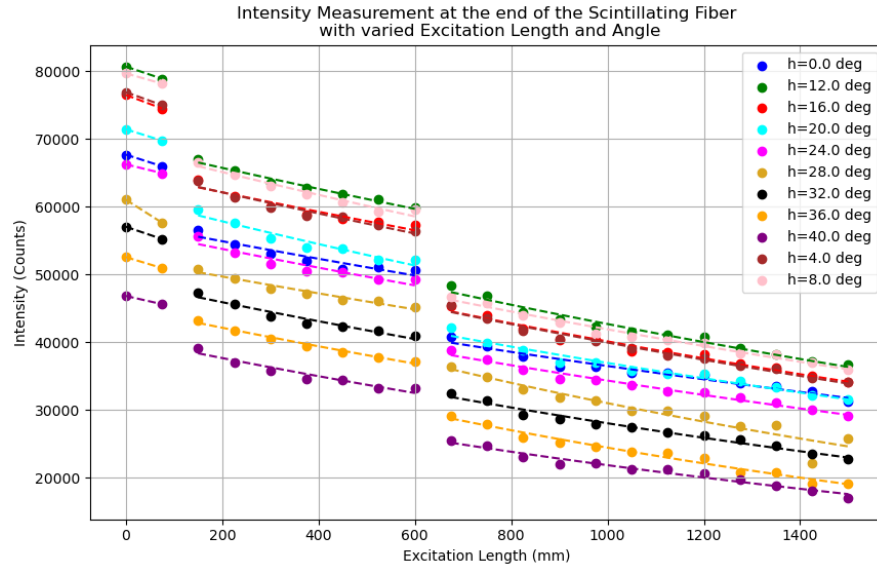


Figure 2.4: Excitation Length and spectrometer angle dependent measurement of intensity at the end of the fibre

2.4 Simulation Analysis

The simulation consists of a dataframe from the GEANT4 software with 12194 simulated data points, each having 15 features. They are simulations of photons propagating down the fiber and getting registered at the photomultiplier end with a given momentum, position and angle with respect to the fiber center axis as well as some valuable attributes.

2.4.1 Un-physical photons

There are events in this dataset where the point at which the photons are supposed to be registered fall outside the width of the excited fiber of 0.125mm. We consider these points as mistakes in the simulation and remove them before proceeding. The removal is done by calculating the exit position vector of the photon and removing all data point with an exit radius larger than 0.125mm as can be seen in [Figure 2.5](#)

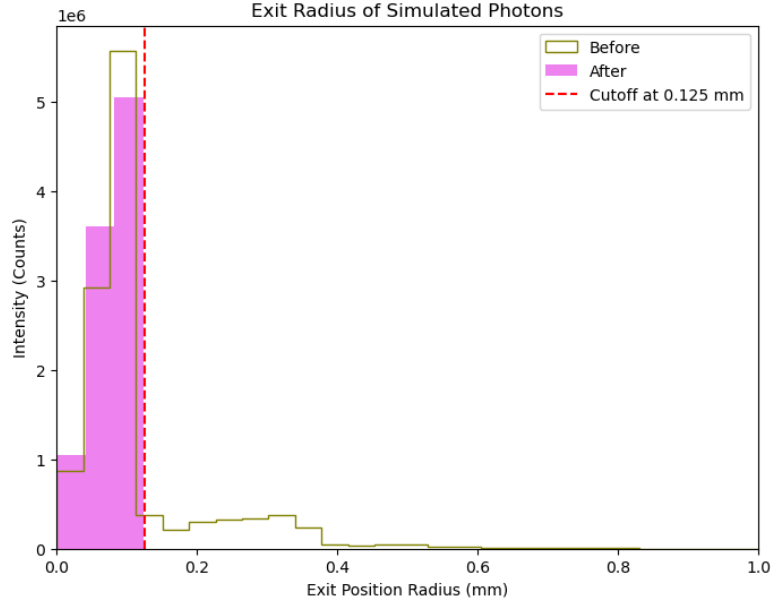


Figure 2.5: Removal of photons with exit coordinates that fall above the width of the scintillating fiber

Another attribute we can use for selecting physical events is to utilize the angle at which the photons arrive at the end of the scintillating fiber. From the internal structure of the fiber, we know that the maximum angles for which internal reflection occurs can be determined with the following forms of Snell's Law:

$$\begin{aligned}
 \theta_1 &= \arccos\left(\frac{n_2}{n_1}\right) = 21.37^\circ \text{ for core-cladding and} \\
 \theta_2 &= \arccos\left(\frac{n_3}{n_1}\right) = 27.44^\circ \text{ for cladding-cladding reflected photons}
 \end{aligned}
 \tag{2.1}$$

Where n 's denote the refractive indexes of the mediums as shown in [Figure 1.3](#). Which means any photons incident on the fiber end with a higher angle must be unphysical as well. Another attribute of the simulation dataset is that it tags photons based on whether they are coming from a core-cladding or cladding-cladding interface reflection, therefore allowing us to separate the dataset into two and apply two separate cuts based on the reflection origin of the photon as shown in [Figure 2.6](#).

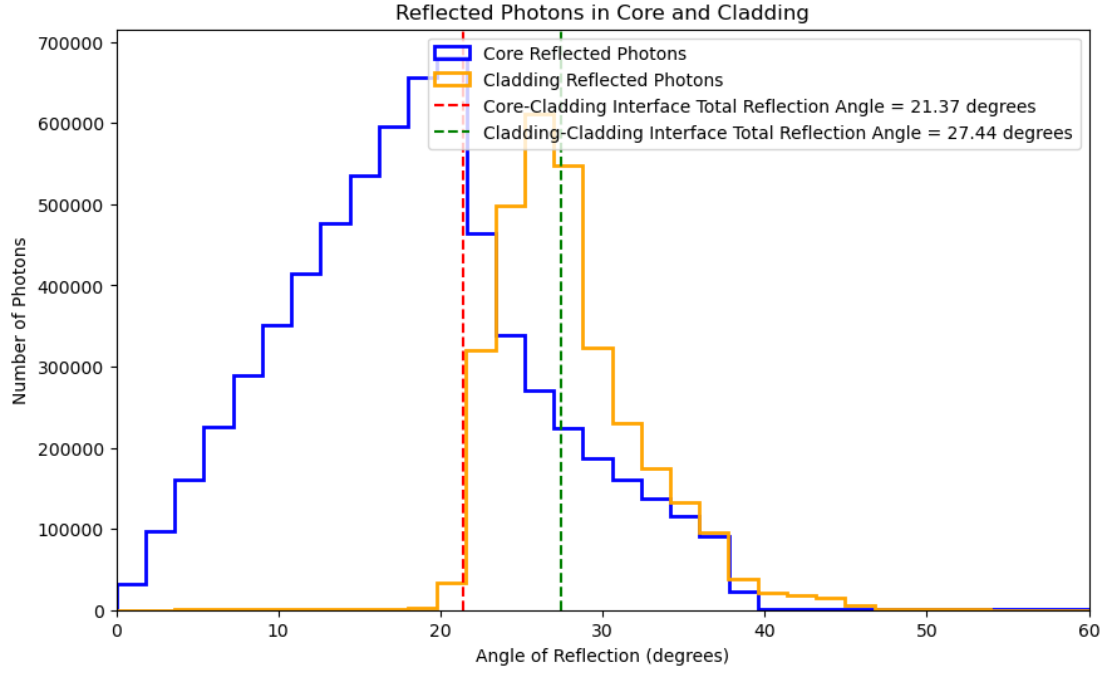


Figure 2.6: Removal of photons with reflection angles higher than physically allowed

2.4.2 Minimum distance (r_{min})

The minimum distance between the path of the photon and the center of the fiber axis (x -axis), as seen in Figure 1.4, is given by the formula:

$$(2.2) \quad r_{min} = \left| \frac{zp_y - p_z y}{\sqrt{p_z^2 + p_y^2}} \right|$$

Where y, z, p_y , and p_z are the positions and momenta of the photon at its creation point in the fiber. This formula is obtained by calculating the minimal distance of the photons path (starting point x, y, z and direction p_x, p_y, p_z) from the x -axis. In order to visualize the connection between this quantity and the reflection angle, we can plot the following 2-D histograms for core-cladding and cladding-cladding photons respectively.

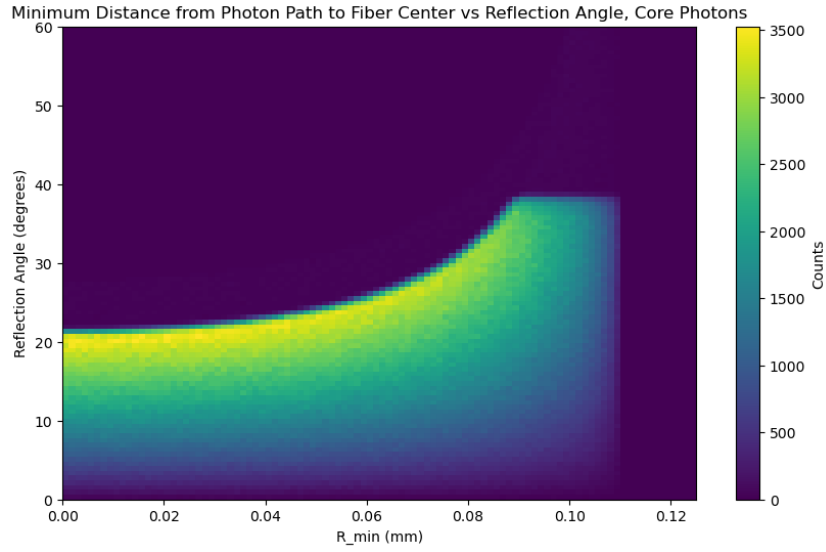


Figure 2.7: 2-D histogram of reflection angle and r_{min} , for core photons

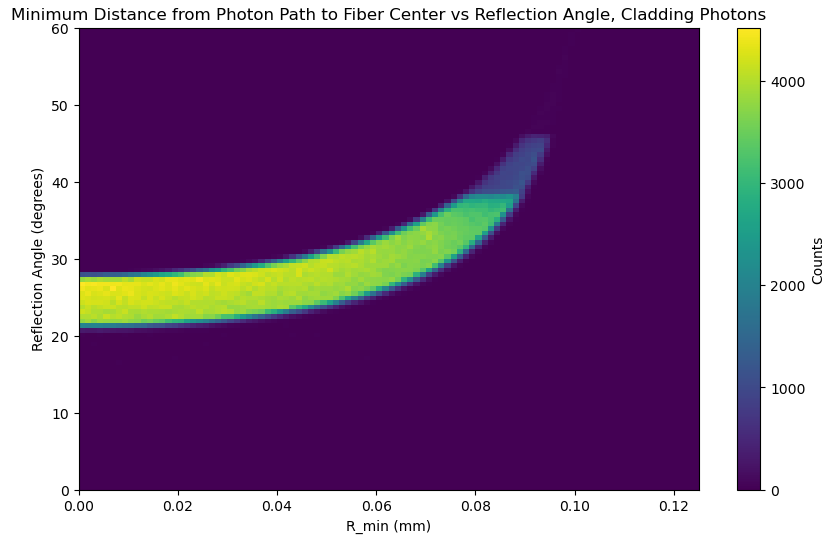


Figure 2.8: 2-D histogram of reflection angle and r_{min} , for cladding photons

We can see that as the minimum distance from the centre of the fiber axis increases, the reflection angle is expected to be higher where the photon ends up following a helical trajectory down the fiber structure. Also of note is that any angle larger than the maximal angle will result in total reflection, hence the distribution seen in both histograms. The cladding photons specifically have to interact with 3 interfaces and hence are not retained below a certain reflection angle.

2.4.3 Attenuation

The final analysis to be done on the simulation data is to illustrate the attenuation, or the intensity of photons that reach the end of the fiber with varying reflection angles and excitation length. The following histogram shows the relation between the two parameters discussed in the intensity formula [Equation 1.8](#) for core-cladding and cladding-cladding photons respectively.

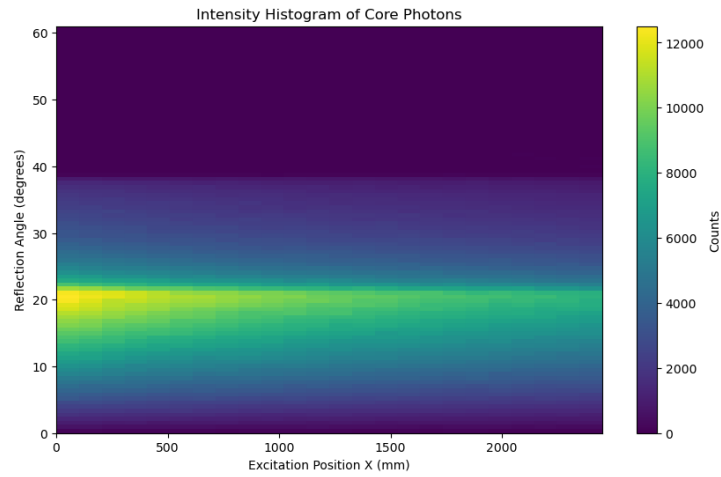


Figure 2.9: Intensity of core-cladding interface reflected photons, plotted as a 2-D histogram with reflection angle and excitation position attributes

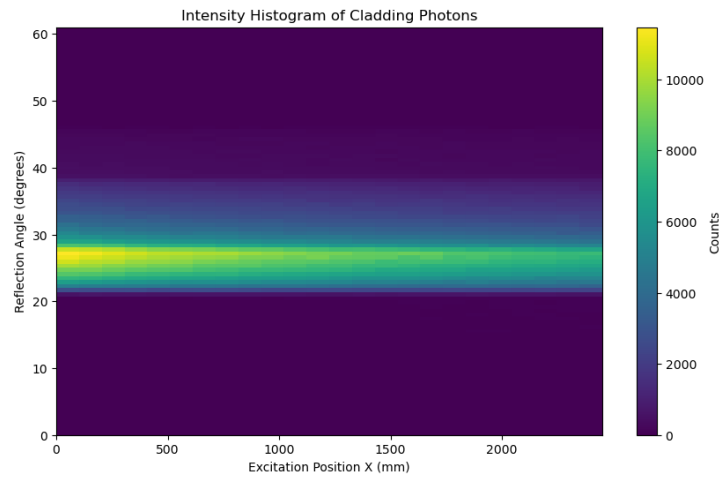


Figure 2.10: Intensity of cladding-cladding interface reflected photons, plotted as a 2-D histogram with reflection angle and excitation position attributes

Using the counts from the histogram bins, we performed curve fits of the intensity in relation to all the increments of excitation length. The intensity function used in the fits will be the following from [Section 1](#):

$$(1.3) \quad I(x) = I_0 \exp(-aL)$$

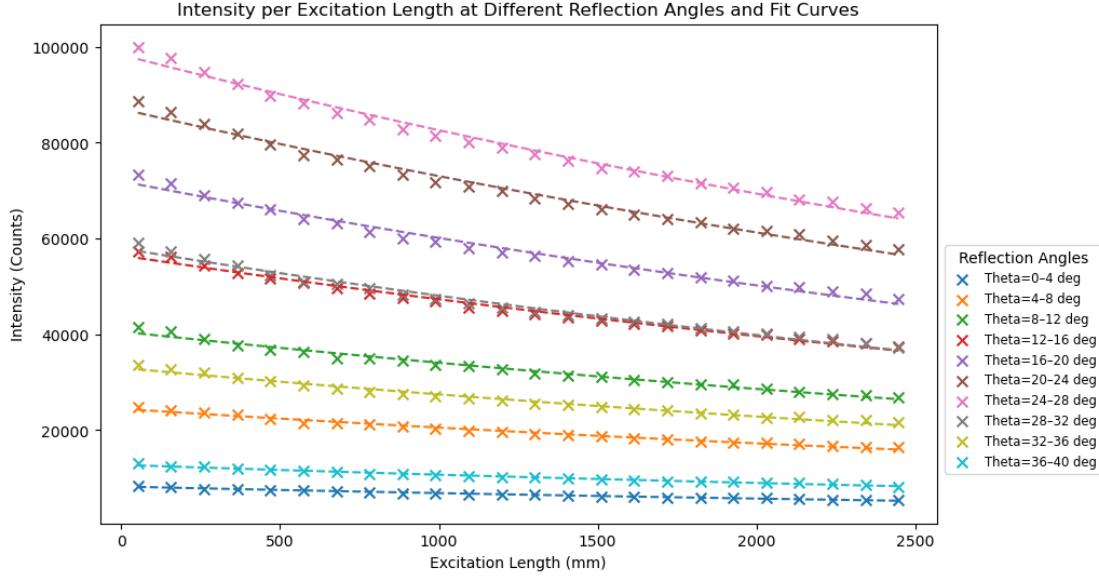


Figure 2.11: Simulation and curve fits for intensity at the end of the fiber for different reflection angles and excitation lengths

Moreover, we can compare the three different ways it is possible for us to acquire the a_{eff} parameter and test our analysis accuracy. For the simulation analysis part, we are going to extract the term from the exponential decay fit in the [Figure 2.11](#). For a theoretical curve, we are going to average the parameters acquired from simulation analysis and make a continuous curve by fitting to the following equation:

$$(2.3) \quad a(\theta) = \frac{A}{\cos(\theta)} + B \tan(\theta),$$

And lastly, we are going to exponential decay fit our data points from the [Figure 2.4](#), where the data comes from our measurement[?]. Plotting the results from the three approaches gives us the following parameter distribution:

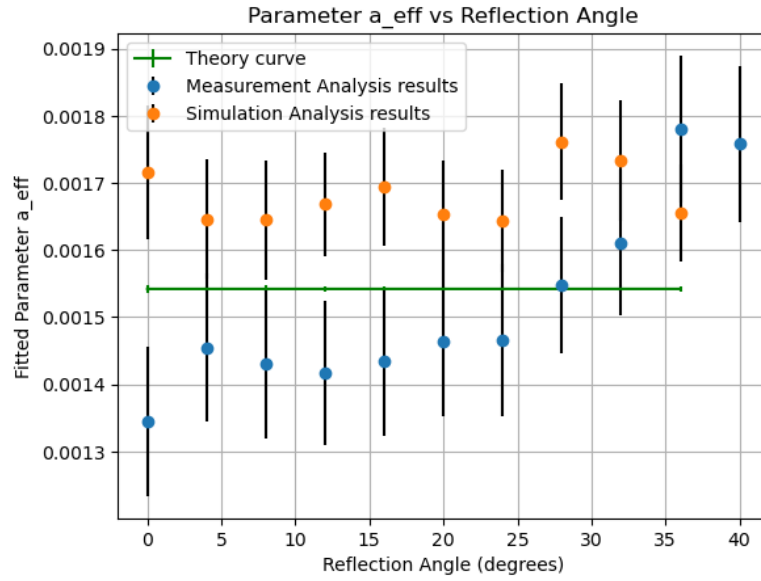


Figure 2.12: Estimation of the inverse attenuation length parameter, the a_{eff} , with respect to the reflection angle by all means possible in this lab course

We have a peculiar result, the theory curve sits almost flat in between the two data distributions that seem to be mirroring each other by the axis of the theory curve between $[0,25]$ degrees angle. Although, the general values are in the same ballpark with the added errorbar.

Conclusions

The roomlight dependent measurements show our spectrometer is resolving different wavelight photons very well. Although the roomlight source is highly dominating in our readings, after subtracting the dark counts, the LED signal was still visible when the roomlight was turned on. We did still proceed to turn off the roomlight for a cleaner signal. Our radial symmetry measurements did give us a symmetric picture, however the center of the distribution was shifted by some degrees instead of being dead center at 0 degrees. This is probably due to misaligned spectrometer arms.

The final measurement, which is the intensity measurement, gave predictable results of an exponentially decreasing intensity as we increase the distance of excitation from the fiber end. Only surprise was from what we understand are a few kinks in the fiber itself which causes a sudden drop in intensity in some LED movement intervals.

When it comes to simulation, we had to remove unphysical data points, however what we removed were always small compared to the whole dataset. The maximum reflection angles were visible in the angular distribution of the simulation data and were in agreement with the calculated ones. The intensity of photons seemed to increase linearly with increased angle until the geometrically allowed limit is reached. This is due to the spherical nature of the photon creation process, photons are radiated isotropically.

Very clear results for the geometrical analysis can be observed in [Figure 2.7](#) and [Figure 2.8](#). The distributions show that the cladding photons have an allowed range in which they can exist with relation to their minimal distance from the center axis and are cut off at an angle of about 21° . On the other hand, the core photons are found where cladding photons are absent. As the reflection angle increases the $r_{\min}$ value also increases for the photon to still propagate in an allowed trajectory of total reflection. This behavior draws a helical shape path inside the fiber in our theory.

The biggest inconsistency with this laboratory report can be observed in the determination of the angular dependence of the fit parameter a . For any of the data samples,

the fit parameter does not show clear dependence on the reflection angle. Moreover, for the theoretical curve there seems to be no dependence at all. As if that was not confusing enough, we observe that when the theory curve is taken as a symmetry axis, the measurement results and simulation results mirror each other. Although, the error allowance added from the evaluation of the fitting parameters allow the values to have crossover at some points. This is the final result of our report.